INCH-POUND

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MILITARY SPECIFICATION

STUD WELDING UNITS, INDEPENDENT DC POWER SOURCE WITH SEPARATE CONTROL UNIT, ELECTRIC ARC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers independent, direct current (DC) welding power sources and separate control units designed for electric arc stud welding with equipment and accessories specified herein. Integral power source and control unit stud welding systems (electric arc and capacitor discharge) are covered by MIL-W-23680. All requirements specified in this document are the minimum requirements of the Government and are not intended to limit the manufacturer in machine design or performance.

1.2 Classification. The power source shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 6.2).

Type I - Transformer Rectifier

Size	Ampere rating	Stud diameter range (inches)
1	1800	12 gauge through 7/8 inch
2	2400	5/16 inch through 1 inch
3	3000	12 gauge through 1-1/4 inches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type III - Diesel Engine Driven Generator

Size	Ampere rating	Stud diameter range (inches)
1	2000	3/8 inch through 7/8 inch

Type IV - Battery Unit

Size	Ampere rating	Stud diameter range (inches)
1	800	10 gauge through 1/2 inch

(Note: Separate control units can be procured without a power source using this specification. See paragraph 3.9.)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MS-51336 - Lunette, Coupler, Drawbar, Ring.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issue of the document which is DOD adopted is the issue listed in the issue of the DODISS cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z49.1 - Safety in Welding and Cutting. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete power source shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The power source shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The power source (when operated with a controller, welding gun, and cables) shall weld studs to the surface of ferrous and nonferrous materials. Welds shall be made without deterioration of the stud or base metal. Stud sizes specified for each system shall be the

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weld base diameter of the stud. The welding unit shall withstand rough handling encountered in construction and fabrication work. The power source shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Reclaimed materials. The welding unit may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the unit. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured products are allowed under this specification.

3.2.2 Energy efficiency. All components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry as provided on the manufacturer's standard commercial product.

3.2.3 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the welding unit, whether or not specifically required herein, shall comply with all requirements of 29 CFR 1910 and ANSI Z49.1. If a conflict arises between 29 CFR 1910 and ANSI Z49.1, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Safety devices normally furnished as standard on the manufacturer's commercial power source supplied to the commercial market shall be provided for the point of operation and all other parts of the stud welding unit that present safety hazards.

3.2.4 Mercury restriction. The welding unit shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.5 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding unit.

3.2.6 Environmental protection. During the construction of the power source and related components, the manufacturer shall comply with all applicable Federal, state, and local statutes which regulate the use of materials hazardous to the ecological system in effect at the point of manufacture on the date of the contract. During the operation of the power source at its final location, it shall not emit materials hazardous to the environment as prohibited by Federal statutes.

3.2.7 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts shall be interchangeable without requiring modification. Accessories normally supplied as standard items shall also be considered replacement parts. Replacement parts shall be domestically available.

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3.3 Construction. The welding unit shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the power source to its maximum rating and capacity without permanent distortion.

3.3.1 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.2 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.4 Painting. Unless otherwise specified (see 6.2), the welding unit shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.5 Threads. All threaded parts used on the welding unit and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4 Type I transformer rectifier power source. Type I power sources shall be solid state, DC, transformer-rectifier power sources. The power source shall be complete with ventilation, adjustable current controls, and a power on-off switch with pilot light. The power source shall have automatically regulated current output to compensate for incoming line fluctuation. Size 1 power sources shall have a maximum current output of not less than 1800 amperes at 16 percent duty cycle and shall be capable of supplying direct current to weld studs 12 gauge through 7/8 inch in diameter. Size 2 power sources shall have a maximum current output of not less than 2400 amperes at 25 percent duty cycle and shall be capable of supplying direct current to weld studs 5/16 inch through 1 inch in diameter. Size 3 power sources shall have a maximum current output of not less than 3000 amperes at 25 percent duty cycle and shall be capable of supplying direct current to weld studs 12 gauge through 1-1/4 inches in diameter. Unless otherwise specified (see 6.2), the power source shall operate from 230/460 volts, 60 Hertz (Hz), 3-phase alternating current (AC) input power and shall be initially wired for operation on 460 volts.

3.4.1 Transformer. The transformer on type I power sources shall produce a continuous current which, when rectified, shall have a direct current rating as necessary to meet the requirements herein. The transformer shall be of the laminated core or shell type. The transformer secondary shall be insulated from the primary and shall not be of the oil-filled type. Coils shall be provided with insulation between the windings, leads, core, and enclosure. Coils shall be protected from moisture. The transformer shall be rigidly mounted within the enclosure. All terminals shall be permanently marked.

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3.4.2 Rectifier. A full wave semiconductor rectifier shall be provided on type I power sources. All components shall be of suitable size and design for the intended service. Surge suppression protection shall be provided with capacity to reduce any transient voltage peaks which would cause component failure. Components subject to failure shall be easily removable to permit replacement.

3.4.3 Insulation. The class of insulation system used shall permit continuous transformer operation and performance at the duty cycle specified herein.

3.4.4 Ventilation system. A forced air ventilation system shall be furnished to minimize internal enclosure heating of the system and allow all components to meet the performance requirements herein. Forced air cooling shall be provided by a totally enclosed, prelubricated, ball bearing fan motor(s). The fan assembly shall be mounted to withstand vibration. Fan blades shall be of a material and design that will prevent fatigue failure under conditions for which the system is designed.

3.4.5 Automatic failure protection. On type I power sources, the primary line voltage feeding the system shall be automatically removed in the event of an overload, excessive weld cycle time, excessive temperature rise, or component failure. All control circuits shall be protected by fuses or circuit breakers. All fuses and circuit breakers shall be accessible for replacement or manual resetting.

3.4.6 Incoming AC power generating set. When specified (see 6.2), a diesel engine driven AC generating set shall be provided to supply incoming power to the stud welding system for remote job site welding where commercially supplied AC power is limited or not available. The generating set shall be rated for the power and duty cycle specified by the procuring activity (see 6.2).

3.4.6.1 AC generating set portability. The AC generating set shall be skid mounted or trailer mounted as specified (see 6.2). When the generator set is be mounted on a trailer, the trailer shall be the manufacturer's standard commercial trailer unless otherwise specified (see 6.2). The trailer shall meet all applicable interstate safety requirements for highway towing. The trailer shall be furnished with the manufacturer's standard commercial ball-type hitch or steel lunette-type ring conforming to MS-51336, as specified (see 6.2).

3.5 Type III engine driven DC generator power source. The type III power source shall be of the diesel engine driven DC generator design. The type III power source shall be rated for not less than 2000 amperes at 15 percent duty cycle, capable of supplying direct current to weld studs 3/8 inch through 7/8 inch in diameter.

3.5.1 Diesel engine. The engine shall be a commercial, industrial type, electric cranking, internal combustion, two- or four-cycle diesel engine having the horsepower, torque, and speed characteristics to meet all performance requirements specified herein. The engine shall operate on #2 grade diesel fuel. The engine shall start and operate in any ambient temperature from 0° to 110° Fahrenheit (F).

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3.5.1.1 Engine controls. The engine shall be provided with an electric start; throttle control; water temperature and oil pressure gauges or warning lights; battery charging gauges; and an emergency stop. The electrical charging system shall have an ammeter or warning light. The engine shall also be furnished with an overtemperature control to shut the engine down in the event of excessive temperature; low oil pressure control to shut the engine down in the event of a low oil pressure condition; and an overspeed control to shut the engine down in the event of excessive operating speed. The engine controls and instruments shall be grouped together on a separate panel or on the welding generator control panel. When specified (see 6.2), an hour meter with a range of 0 to 99,999 hours in increments of not greater than 1 hour shall be provided with the engine.

3.5.1.2 Battery. Unless otherwise specified (see 6.2), batteries required for operation shall be provided. The batteries shall be of the maintenance-free type.

3.5.1.3 Low temperature starting aid. When specified (see 6.2), the engine shall have a cold weather starting device. Necessary warning statements as to the use of starting aid fluid shall be printed on a corrosion resistant plate secured near the starting aid control.

3.5.1.4 Governor. A governor shall be furnished and set to limit the engine speed to the manufacturer's recommended operating speed.

3.5.1.5 Pollution control. The engine shall meet all applicable Environmental Protection Agency emission control regulations in effect on the date of the contract. The engine shall not exceed OSHA noise level requirements during its operation.

3.5.1.6 Cooling system. The engine shall be provided with thermostat controlled liquid cooling system of adequate size and design to operate at an ambient temperature of 110°F.

3.5.1.7 Fuel system. A fuel tank with at least a 65 gallon capacity shall be provided. A low fuel level shutdown device shall be provided to prevent the engine from running out of fuel. The fuel system shall be provided with a strainer to strain sediment from the fuel and a replacement cartridge type filter located between the strainer and fuel pump.

3.5.2 Generator. The generator shall be the direct current, self-regulating type, complete with weld voltage control, a weld volt meter, engine tachometer, and output terminals. The maximum temperature rise of the components shall not affect generator operation and performance at the duty cycle specified herein.

3.6 Type IV battery unit power source. Type IV battery unit power sources shall be comprised of a series of batteries complete with a battery level meter, charging indicator light, fuse, and power on-off switch with pilot light. The unit shall be self-contained with built-in charging and control circuitry. The unit shall charge from 110 volts, 60 Hz, single phase input AC power and shall switch off when full charge is reached. The type IV power source shall be rated at 800 amperes. The type IV power source shall supply direct current to weld studs 10 gauge through 1/2 inch in diameter.

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3.7 Enclosure. Each power source shall be encased in a ventilated, dripproof metal enclosure with the strength and rigidity necessary to resist the abuse encountered in a stud welding environment without increasing the risk of fire, electric shock, or operator injury. Metal parts shall be protected from corrosion. Access openings shall be easily removable for inspection, cleaning, and servicing. All openings shall be protected to prevent the entrance of trash or rodents.

3.8 Portability and rough handling. The power source shall be portable. The power source shall be: (1) mounted on a base and provided with means for lifting at a designated point by crane or hoisting mechanism equipped with a hook, shackle, or clevis or by a fork lift truck; (2) mounted on a rolling trailer; or (3) mounted on a hand cart with wheels, as specified (see 6.2). The lifting device, when provided, shall be firmly and rigidly attached to the enclosure, capable of supporting 2-1/2 times the unit's weight. Rough handling shall not cause distortion of the frame or any of its components. The mount shall be large enough to accommodate a power source and a controller.

3.8.1 Trailer. When the power source is be mounted on a trailer, the trailer shall be the manufacturer's standard commercial trailer unless otherwise specified (see 6.2). The trailer shall meet all applicable interstate safety requirements for highway towing. The trailer shall be furnished with the manufacturer's standard commercial ball-type hitch or steel lunette-type ring conforming to MS-51336, as specified (see 6.2).

3.9 Stud welding controller. An independent stud welding controller shall be provided when specified (see 6.2). The controller shall be complete with reverse polarity capability (positive or negative ground), polarity switch, solid state time control, and a power on-off pilot light. All controls shall be located on the front of the controller. The controller shall be protected by a fuse. The controller shall have a stud diameter welding range of either 12 gauge through 5/8 inch or 1/4 inch through 1-1/4 inches as specified (see 6.2). The controller must have welding capability which is compatible with the power source specified in 1.2 when the two components are to be used together. When specified (see 6.2), the controller shall be furnished with one or more of the following special control modules: (1) a safety interlock module to allow more than one controller to be operated from a single power source; (2) a chuck saver module to prevent accidental arcing of the chuck; and (3) a plunge dampener to minimize weld spatter and provide greater fillet control.

3.10 Stud welding gun. When specified (see 6.2), a stud welding gun shall be provided. The gun shall be of the portable, pistol type with a finger-actuated switch. The housing shall be made of high-impact plastic with electrical insulating properties. Control and gun power cables shall enter the bottom of the handle or be secured to the bottom of the handle. The gun shall not overheat or wear prematurely when welding studs of the size specified herein when operated in an ambient temperature of 110°F without auxiliary or forced cooling.

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3.10.1 Gun accessories. Gun grips, foot and leg assemblies, and chucks shall be provided as specified by the procuring activity (see 6.2). All gun accessories supplied shall fit the stud gun without requiring modification to the gun or accessory and shall weld all stud diameters required for the type and size unit specified in 1.2.

3.10.2 Gun tools. Each stud gun shall be furnished with a complete set of tools required for its operation and maintenance.

3.11 Welding cables. When specified (see 6.2), the stud welding unit shall be provided with cables. The cables carrying the welding circuit current and the welding circuit ground shall be manufacturer's standard resistant to oil, acid, alkali, and heat. The cables shall be capable of carrying the amperage necessary to weld all stud diameters required of the type and size of unit specified in 1.2. The cables shall have quick disconnect connectors. Unless otherwise specified (see 6.2), not less than a 15 foot control cable, a 15 foot welding cable, and a 15 foot ground cable with clamp shall be provided. Combination control and welding cables can be provided.

3.12 Gas shielding equipment. When specified (see 6.2), a gas flow meter and regulator to control the flow of shielding gas to the stud gun and to measure the cylinder pressure; shielding gas hose suitable for use with all gases used in the metal inert gas shielded and metal gas shielded welding process; a gas spark shield; and all necessary adaptors and connectors for shielded arc welding shall be provided. Additional gas shielding equipment required by the procuring activity shall be as specified and fully described (see 6.2).

3.13 Performance. Each stud welding unit shall be capable of welding studs to meet the requirements in 3.13.1 and 3.13.2.

3.13.1 Weld visual inspection requirement. The fillet formed during the stud welding with the power source units covered by this specification shall be a well-formed fillet, wetted to the base plate. All welds (except for aluminum welds) shall have a bluish cast with no evidence of undercutting, inclusions, porosity, cracks, or excessive spatter.

3.13.2 Weld bend requirement. Five of the studs welded using the power source units covered by this specification shall be hammered over or bent using a bending tool 45° or until the stud or base plate fractures. Failure shall occur in the stud or on the base plate or a plug of base metal shall tear out. There shall be no evidence of failure or cracking in the weld zone.

3.14 Standard equipment. All standard equipment normally furnished with the manufacturer's commercial equipment shall be furnished.

3.15 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.16 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

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3.17 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall permanently marked in boldface on a contrasting background. All controls shall be clearly marked as to their function. All plates shall be corrosion resistant.

3.17.1 Nameplate. A nameplate shall be securely attached to each enclosure. Unless otherwise specified (see 6.2), the nameplate shall provide the following information:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.17.2 Weld cycle chart. Recommended welding times and currents for the stud diameter range applicable to the size power source furnished shall be permanently marked on a chart or plate on the controller when furnished (see 3.9). Weld diameters shall be given in both inches and millimeters.

3.17.3 Diagram of connections. A diagram of electrical connections shall be attached to the inside of each power source enclosure. The diagram shall be protected with transparent varnish, transparent plastic covering, or other transparent material.

3.18 Workmanship. Workmanship of the power source and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall

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not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The unit shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.2.1 and 4.7.2.2, and the inspection in 4.8. The unit shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each power source and all accessories and optional equipment furnished shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 50° to 104°F and 20 percent to 95 percent relative humidity.

4.7.2 Performance tests. Each system shall be capable of performing all tests specified herein as applicable to the type and size unit furnished. If a complete stud welding unit (power source, controller, welding gun, and cables) is not specified for a procurement, a complete welding unit shall be used to perform the test(s). At the option of the procuring activity, each unit shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2). All performance tests shall be completed without performing any maintenance on any unit components.

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4.7.2.1 Operational test. All circuits in the welding unit shall be energized. Proper operation of all controls, motors, adjusting mechanisms, optional equipment, and accessories shall be verified without performing any actual welding.

4.7.2.2 Maximum current output test. The stud welding unit shall be tested to determine that the maximum current output meets the requirements for the power source given in section 3 for the type and size unit furnished.

4.7.2.3 Welding test. Each unit shall be tested to ensure it meets the performance requirements specified herein. Ten round, low carbon steel studs (C-1008 to C-1010) ranging in diameter from 1/8 inch to 7/8 inch shall be welded to a low carbon steel base metal (C-1008 to C-1030). The base metal shall be not less than 1/3 the thickness of the diameter of the stud. All welds shall conform the requirements of 3.13.1 and 3.13.2.

4.7.2.4 Gas shielded weld test. When gas shielding equipment is specified (see 3.12), the equipment shall be tested to ensure it functions properly and meets the performance requirements specified herein. Ten round, aluminum alloy (5356) studs ranging in diameter from 1/8 inch to 1/2 inch shall be welded to an aluminum alloy base metal (1100, 3000, or 5000 series). The base metal shall be not less than 1/3 the thickness of the diameter of the stud. Proper operation of the gas shielding equipment shall be determined. All welds shall pass the inspections in 3.13.1 and 3.13.2.

4.7.3 Type I safety devices.

4.7.3.1 Overtemperature protection test. Each type I power source unit provided with forced air cooling (see 3.4.4) shall be tested for automatic excessive temperature rise failure protection. With the power source operating in a thermally stabilized condition at rated load current and with enclosure panels in place, the fan motor shall be disconnected. The automatic failure protection device shall comply with the requirements of 3.4.5.

4.7.3.2 Automatic failure protection test. An overload condition, excessive temperature rise, excessive weld cycle time, or component failure shall be simulated to test the automatic failure protection device. The protection device shall comply with 3.4.5.

4.7.4 Type III safety devices.

4.7.4.1 Low oil pressure control device. The low oil pressure shutdown device (see 3.5.1.1) shall be tested with the engine running. The pressure to the device shall be reduced without actually reducing the engine oil pressure. Failure of the device to shut down the engine shall be cause for rejection.

4.7.4.2 Overspeed shutdown device. The overspeed safety device (see 3.5.1.1) shall be tested with the engine running. The engine speed shall be manually adjusted to an overspeed setting (higher than the manufacturer's recommended operating speed). Failure of the overspeed device to shutdown the engine shall be cause for rejection.

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4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The equipment covered by this specification is intended to provide a DC power source for the stud welding of carbon steel, stainless steel, and aluminum alloys. Galvanized steel, copper, and brass can also be welded with special provisions. The power source and controller shall be specified as individual components or as a complete stud welding unit. The stud welding units will be operated in indoor and outdoor environments where electric power is not available or portability is desired such as construction and fabrication sites, repair shops, shipyards, aboard ships, and aircraft and ground support equipment sites.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Additional safety and health requirements, if required (see 3.2.3).

f. Painting, if different (see 3.3.4).

g. Electrical requirements on type I power source, if different (see 3.4).

h. AC generating set, if required; specify power and duty cycle (see 3.4.6).

i. Specify AC generating set mounting (3.4.6.1).

j. AC generating set trailer, if different; specify and fully describe or cite applicable military standard for trailer (see 3.4.6.1).

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k. Specify ball-type or lunette ring-type trailer hitch for AC generating set trailer, if required (see 3.4.6.1).

l. Type III hour meter, if required (see 3.5.1.1).

m. Type III starting system batteries, if not required (see 3.5.1.2).

n. Type III engine starting aid, if required (see 3.5.1.3).

o. Type of power source mounting required (see 3.8).

p. Power source trailer, if different; specify and fully describe or cite applicable military standard for trailer (see 3.8.1).

q. Specify ball-type or lunette ring-type trailer hitch for power source trailer, if required (see 3.8.1).

r. Controller, if required; specify stud diameter range required (see 3.9).

s. Specify special controller modules (safety interlock, chuck saver, and/or plunge dampener), if required (see 3.9).

t. Stud weld gun, if required (see 3.10).

u. Gun accessories, if required; specify and fully describe (see 3.10.1).

v. Cables, if required; specify length, if different (see 3.11).

w. Gas shielding equipment, if required; specify additional equipment, if required (see 3.12).

x. Specify and fully describe optional equipment, if required (see 3.15).

y. Specify applicable repair parts, if required (see 3.16).

z. Nameplate, if different (see 3.17.1).

aa. Inspection conditions, if different (see 4.3).

bb. First article inspection, if different (see 4.4).

cc. Quality conformance inspection, if different (see 4.5).

dd. Performance tests, if different (see 4.7.2).

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6.3 Technical data. When technical data is specified on a Contract Data Requirements List (DD Form 1423) in the contract, it should be specified in the "Remarks" section of the DD Form 1423 that all technical data shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-W-80110B	MIL-W-80110C
Type I	Type I, Size 1
New	Type I, Size 2
New	Type I, Size 3
Type III	Type III
Reinstated	Type IV

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6.9 Safety and health requirements. Standard 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

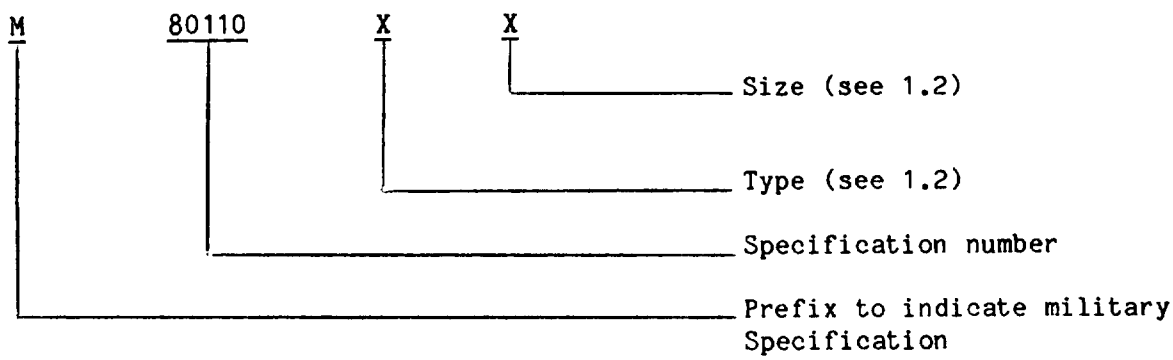
6.11 Training. Training required by the procuring activity should be provided as specified in the contract.

6.12 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.13 Subject term (keyword) listing.

Arc welding
Battery
Diesel engine generator
Joining
Torches
Transformer rectifier

6.14 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



(The Roman numerals designating machine types shall be represented in the PIN using Arabic numbers, i.e., "I" shall be represented as "1".)

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84

User activities:
Navy - AS, MC

Preparing activity:
DLA - IP

Project (3431-0186)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80111C	Stud Weld Units, Independent DC Power Source/Control Unit

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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